

LEVEL DESIGN DOCUMENTATION TEMPLATE (KOTH_CONCRETE)

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High Level Overview

Level Goals

Main Objectives

The main objective of this level design GDD is to create a full-fledged level complete with all the necessary conventions within Team Fortress 2. In addition to having a complete level the level's design should create an engaging experience which considers the gameplay of Team Fortress 2. Ideally, the gameplay within the level should be balanced and not favour any specific playstyle over another. An overview of the level's main objective can be found below.

- A complete level experience which functions for the gamemode of King of the Hill. The complete experience includes necessary functionality within the game such as respawning, capturing objectives and other concepts found within official Valve Maps.
- Level should be enjoyable to play and engaging experience within my level that directs the players towards the gameplay objective.
- The level works for the main playstyles within the game and no specific class is objectively overpowered with each class being offered level design effective for them.
- Balance focus should also be around a 6v6 team composition and considerations that factor in should consider that when relevant.

Secondary Objectives

Alongside the main objectives of the level, there are secondary objective to be considered. These objectives focus on visual aspects and smaller further designs which could implemented within the level such as visuals and additional design.

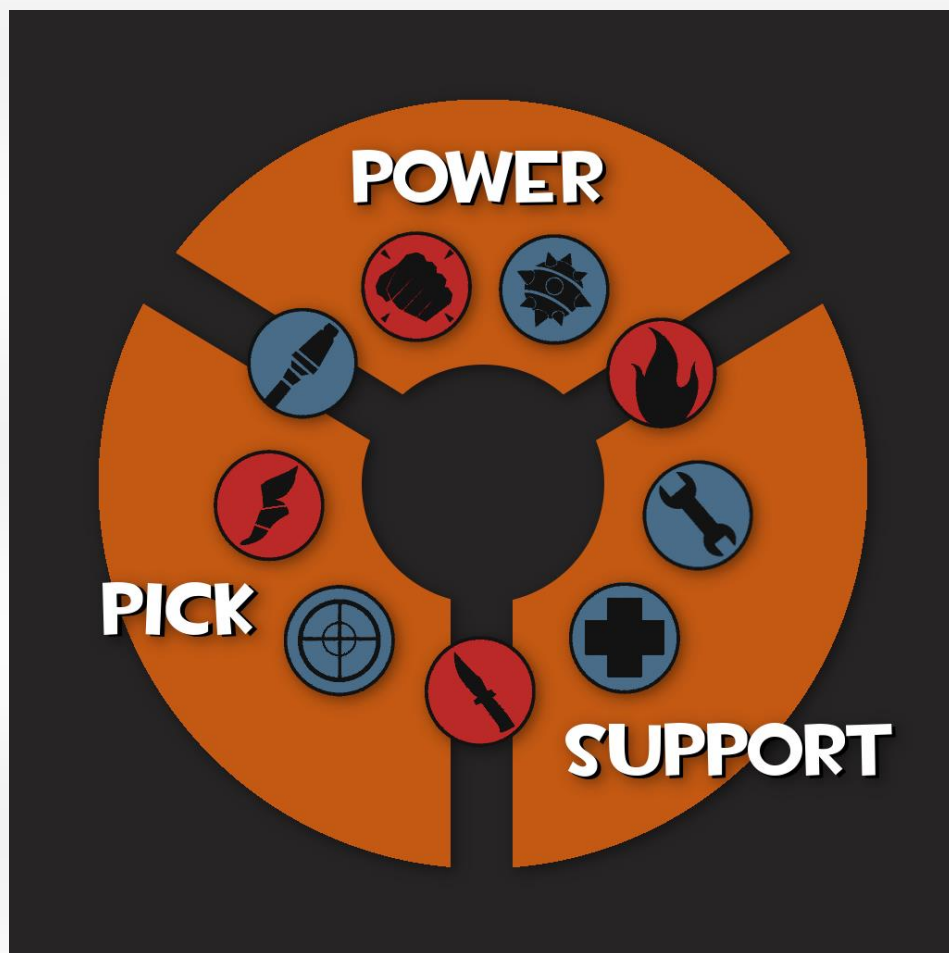
- A visually interesting map which is stylistically consistent with the map pool within the official game.
- Additional level design techniques such as affordances to direct player behaviours.
- A level that is recognisable and unique from the pre-existing map pool with identifiable characteristics that players can use to reference the level from. "The map with [Blank]"
- A level that properly considers each class individually and makes attempts to improve their specific implementations further.



Design Considerations

Playstyle Considerations

Within Team Fortress 2 there are three main playstyles that can be used to group the individual classes within. The groupings are based on the functionality of the classes and are called Pick, Power and Support. Certain classes are purely one of the three, but several derive functionality from two of the functionalities. Grouping the classes into a generalised set of identifiable playstyles helps to create a useful base to develop with. This broad design approach will allow areas to effectively work for all classes, from there further refinement can be added to create additional considerations for specific classes.





Pick

Pick classes are defined by their individual high damage output which is focused on an individual. In addition to that, pick classes typically play their advantage by choosing their fights and are given options in functionality that allow them to achieve this. For example, Scout with his high mobility, Sniper with his infinite range, and Spy with his on-demand invisibility. Areas that are effective for pick classes tend to be focused on allowing them to utilise them to utilise their ability to choose engagements. In this case, those areas tend to be open, long, and covered in smaller cover with flank routes that allow specific pick classes to position themselves in advantageous positions. An example of effective pick class implementation can be found in the map of Lakeside.



Power

Power classes are defined by their ability to process longer and more consistent damage outputs which are effective at wiping out scores of opponents. Pure power classes forfeit the same affordances in picking fights in favour of this strength which can be seen in the class of Heavy and Demoman. These two classes do not have the same effective methods of picking fights found within pick classes. These classes heavily rely on areas where enemies are funnelled into their effective ranges which is how they push their advantage. Power classes considerations would include creating chokepoints for enemies to funnel into and enclosed spaces. An example of which can be found on the map of Lakeside.



Support

Support classes are a smaller role within Team Fortress 2 but are usually niche roles designed to offer support to teams through their functionality. There is no way of identifying a specific set of traits for support roles besides not front lining and assisting the team. Medic and Engineer are examples as they prefer to play behind the team as well as offering support in the form of healing. Effective implementation within a level



for supports if offering effective cover nearby key areas that teams would be assaulting from. In addition to that offering effective map cover throughout is useful for their safety and to effectively operate.



Class Considerations

Alongside the roles, individual considerations must be made for each class. Targeting specific classes in this way would allow the level to improve the experience for the player. Class considerations are smaller considerations within a level such as inclusion of additional designs that support the specific characters' interactions.

Scout

Open areas for duelling alongside effective flank routes allow scout to effectively play. His on-demand mobility and an area to move allow him to duel effectively as the space allows him to dodge. Additionally, flank routes allow him to reposition behind enemies and utilise his ability to pick fights. To further implement interesting movement for the Scout implementation of elevated positions that he can quickly jump between using his additional jump should be included.



CLASS - SCOUT



CLASS STATS

HEALTH - 125 HP [185 OVERHEAL]

SPEED - 133% [400 Hu/s]



CLASS INFORMATION

THE QUICKEST CLASS IN THE GAME, HAS THE ABILITY TO DOUBLE-JUMP AND ACCOUNTS FOR TWO PLAYERS ON OBJECTIVES. THE CLASS HAS ACCESS TO A HIGH-DAMAGE, CLOSE-RANGED SHOTGUN WHICH DOES CONSIDERABLE SINGLE TARGET DAMAGE IN THE EFFECTIVE RANGE. PLAYSTYLE FOCUSES ON PICKING ENEMIES IN DUELS, FLANKING AND UTILISING THE ON-DEMAND MOVEMENT TO PICK FIGHTS.

LOADOUT



TO CATER THE LEVEL TO THIS CLASS, IMPLEMENTATIONS OF FLANKS AND OPEN AREAS WHERE HIS MOVEMENT CAN BE EFFECTIVE. ADDITIONALLY, SMALLER ELEVATION THAT HE CAN QUICKLY JUMP AND UTILISE IN COMBAT WILL BOOST HIS EFFECTIVENESS.

Soldier

Soldier is a very powerful and well-rounded class that struggles naught to find an effective position within any level. The ability to travel quickly alongside a high, easy-to-use damage output with a sizeable health pool means there isn't much in the way that can negatively affect Soldier. The best way to add considerations for this class is add healthpacks alongside typical rocket jump pathways usually nearby objectives.



Additionally, avoid having low ceilings as they encourage soldier to use his high-skill and engaging rocket jumping ability.

CLASS - SOLDIER



CLASS STATS
HEALTH - 200 HP [300 OVERHEAL]
SPEED - 80% [240 HU/s]



CLASS INFORMATION
A GENERALIST WITH POWERFUL MOBILITY AND SIZEABLE HEALTHPOOL. DESPITE A LOW GROUND SPEED THE ABILITY TO UTILISE ROCKETS ALLOWS FOR SUBSTANTIAL MOBILITY. HIS HIGH-DAMAGE, SPLASH DAMAGE ROCKETS, AND SHOTGUN ALLOW SOLDIER TO EFFECTIVELY DEAL WITH ANY THREAT.

TO CATER THE LEVEL TO THIS CLASS, IMPLEMENTATIONS OF FLANKS, ENCLOSED SPACE AS WELL AS ELEVATION AND RAMPS BENEFIT SOLDIER'S GAMEPLAY. FLANKS, ELEVATION AND RAMPS ALLOW SOLDIER TO UTILISE ROCKET JUMPING TO EFFECTIVELY POSITION. ENCLOSED SPACES ALLOW FOR SOLDIER TO EFFECTIVELY UTILISE THE SPLASH DAMAGE FROM ROCKETS. ADDITIONALLY, HEALTHPACKS ACROSS AREAS WHERE ROCKET JUMPING SOLDIERS ROUTES SUPPORT THE CLASS.

LOADOUT


Pyro

Pyro is a close-range ambush class that aims to quickly engage enemies around compact spaces. The overall inability to deal with enemies from afar can be a weakness for this class. A good way of implementing considerations for the character is to offer space where that is compact and has corners to ambush from. Pyro also benefits from death pits as their airblast can be used to knock enemies to instant-kill hazards.



CLASS - PYRO


CLASS STATS
HEALTH - 175 HP [260 OVERHEAL]
SPEED - 100% [300 Hu/s]


CLASS INFORMATION
A CLOSE-RANGE SPECIALIST. PYRO WIELDS A VARIETY OF FIRE-BASED OPTIONS. WITH SUCH LIMITED RANGED OPTIONS PYRO RELIES ON AMBUSHING ENEMIES AND TAKING ALTERNATIVE ROUTES TO ACHIEVE THIS. ADDITIONALLY, PYRO HAS THE ABILITY TO REFLECT MOST PROJECTILES USING HIS COMPRESSION BLAST. THE COMPRESSION BLAST CAN ALSO BE USED TO EXTINGUISH TEAMMATES AND PUSH PLAYERS. THE CLASS HAS LIMITED MOBILITY OPTIONS SUCH AS THE THERMAL THRUSTER AND DETONATOR.
LOADOUT

TO CATER THE LEVEL TO THIS CLASS, IMPLEMENTATIONS OF FLANKS, ENCLOSED SPACE WITH PLENTY OF CORNERS TO ALLOW FOR AMBUSHES. ADDITIONALLY, CLIFFS AND LETHAL HAZARDS ENABLE THE CLASS TO PUSH PLAYERS INTO INSTANT DEATH WHICH IS GOOD FOR PYRO.

Demoman

Focusing on raw damage output, Demoman doesn't struggle to fit within a level if considerations made for power classes are implemented well. Considerations that could be made for Demoman specifically can come in the form of inclusion of doorways that can be sticky trapped.



CLASS - DEMOMAN



CLASS STATS

HEALTH - 175 HP [260 OVERHEAL]

SPEED - 93% [280 Hu/s]

CLASS INFORMATION

A POWERFUL CLASS THAT WREAKS HAVOC ON ENEMIES ENCAMPMENTS AND BUILDINGS. THE CLASS FOCUSES ON MAINTAINING DISTANCE, CONTROLLING SPACE AND DEFENDING OBJECTIVES. THE CLASS STRUGGLES IN WITH ENEMIES IN CLOSE RANGE AND OPEN AREAS. MOBILITY IS LIMITED, BUT HAS THE ABILITY TO STICKY-JUMP. ADDITIONALLY, DEMOMAN IS ABLE TO SWAP HIS LOADOUT TO A MELEE FOCUSED CLASS WITH INCREASED STATS.

TO CATER THE LEVEL TO THIS CLASS, IMPLEMENTATIONS OF ENCLOSED SPACES, CHOKEPOINTS AND OBJECTIVES ALLOW DEMOMAN TO UTILISE HIS STRENGTHS. ADDITIONALLY, DOORWAYS ALLOW FOR DEMOMAN TO SET UP STICKYBOMB TRAPS AND AMBUSH ENEMIES

LOADOUT



Heavy

Just as Demoman, considerations for Heavy are effectively made once you have effective implementation for Power classes. Being so vulnerable to sniping at their low speed, having decent cover on areas Heavy would like to hold can help make the class less vulnerable.



CLASS - HEAVY



CLASS STATS

HEALTH - 300 HP [450 OVERHEAL]
SPEED - 77% [37] [230 [110] Hu/s]



CLASS INFORMATION

A POWERFUL CLASS THAT SHREADS ENEMIES WITH HIS MINIGUN. THE CLASS IS DEFINED BY THE LARGEST HEALTHPOOL, HIGHEST CONSISTENT DAMAGE AND SLOWEST SPEED OF ANY CLASS. SPEED THAT GOES EVEN LOWER WHEN USING HIS MINIGUN. THE LACK OF ANY MOBILITY MAKES HEAVY A HUGE TARGET FOR SNIPERS.

TO CATER THE LEVEL TO THIS CLASS, IMPLEMENTATIONS OF MODERATELY-SIZED AREAS THAT ENEMIES ARE FUNNELED INTO. GOOD COVER FROM SIGHTLINES SUPPORT HEAVY AGAINST SNIPERS.

LOADOUT



Engineer

Engineer has only two main considerations, effective building placement and ammo packs. A good position for a sentry is important as the engineer heavily relies on his equipment and should be placed in a safe, position behind the frontline of combat. Dispensers should have places for cover near and around combat zones such as objectives. Good positions for teleporters are in areas that allow the engineer nearby



the objective but out the way and behind cover. Additionally, ammo packs are necessary to include on a map because engineer relies on them to resupply metal on the frontline, so considerations for them should be made.

CLASS - ENGINEER

LOADOUT


CLASS STATS
HEALTH - 125 HP [185 OVERHEAL]
SPEED - 100% [300 Hu/s]

CLASS INFORMATION
A DEFENSIVE SUPPORTIVE CLASSES. THE ENGINEER HAS A VARIETY OF TOOLS TO SUPPORT HIS TEAM. TYPICALLY PLAYING BEHIND YOUR TEAM, THE ENGINEER UTILISES HIS DISPENSER AND TELEPORT TO SUPPLY AND TRANSPORT TEAMMATES ON THE FRONTLINE. HIS SENTRY IS A GREAT AREA DENIAL TOOL THAT QUICKLY KILLS ENEMIES. THE CLASS HAS LIMITED MOBILITY AND A SMALL HEALTH POOL WITH WEAPONS THAT ENABLE THE CLASS TO BECOME MORE OR LESS OFFENSIVE.
TO CATER THE LEVEL TO THIS CLASS, IMPLEMENTATIONS OF AREAS THAT ELEVATED FOR SENTRIES ALONGSIDE PLENTY OF AMMO PICKUPS HELP. ROOMS AND COVER NEAR THE OBJECTIVE FOR TELEPORTERS AND DISPENSERS.

Medic

Medic has no specific considerations for their class implementation within a level, besides cover. Considerations for sightlines and generous cover will make the high-priority class less vulnerable. Additionally, too many opportunities included such as an overabundance of flank routes will negatively affect the class and is something that has to be considered.



CLASS - MEDIC


CLASS STATS
HEALTH - 150 HP [225 OVERHEAL]
SPEED - 107% [320 Hu/s]


CLASS INFORMATION
A FULL-SUPPORT CLASS, THE MEDIC IS THE TEAM'S DEDICATED HEALER ROLE. THE ROLE OF MEDIC IS TO MAINTAIN THE TEAM'S HEALTH BY UTILISING HIS MEDI GUN TO REGENERATE HITPOINTS. MEDIC HAS THE ADDITIONAL ABILITY TO OVERHEAL TEAMMATES UP TO 150% THEIR BASE HEALTH POOL. THE CLASS HAS A SLIGHTLY FASTER BASE SPEED, LOW MOBILITY AND IS OFTEN THE TARGET OF ENEMIES. THE CLASS NATURALLY REGENERATES 3-6 HP A SECOND SCALING UP BASED ON THE TIME SINCE LAST HIT.
LOADOUT

TO CATER THE LEVEL TO THIS CLASS, IMPLEMENTATIONS OF EFFECTIVE STRONG COVER AND ROUTES THAT AVOID MAJOR SIGHTLINES HELP. TOO MANY FLANKS CAN NEGATIVELY IMPACT CLASS SURVIVABILITY SINCE THE CHARACTER IS A HIGH-PRIORITY PICK.

Sniper

Sniper only has an interest in long sightlines. Good positions where sniper can hold that are relatively safe and cover a fair amount of distance are a very important consideration for the class. Since his range is effectively infinite, sightlines need to be balanced to not allow him to dominate over an entire map. But this of course should



not make the class entirely unviable and proper testing of his viability should be done in game.

CLASS - SNIPER


CLASS STATS
HEALTH - 125 HP [185 OVERHEAL]
SPEED - 100% [300 Hu/s]



CLASS INFORMATION
A DEDICATED-RANGED CLASS SNIPER IS A MEDIOCRE CLASS BALANCED AROUND HIS INFINITE RANGE. HIS ROLE IS TO STAY BEHIND THE FRONTLINE AND UTILISE HIS RANGE TO PICK OFF SPECIFIC TARGETS. WITH NO MOBILITY OPTIONS AND LIMITED COMBAT VIABILITY AT CLOSE RANGE THE CHARACTER'S POWER COMES FROM THE ABILITY TO INSTANTLY KILL ALL CLASSES FROM ANY RANGE.

TO CATER THE LEVEL TO THIS CLASS IS TO CONSIDER THE VIABILITY OF IT'S IMPLEMENTATION. SIGHTLINES NEED TO BE BALANCED. NOT TOO STRONG THAT HE CAN SHUTDOWN THE ENTIRE MAP, BUT ENOUGH SPACE SO THE CLASS ISN'T UNVIALE. IT'S UNIQUE IN THAT YOU NEED TO CONSIDER GEOMETRY TO REDUCE THE EFFECTIVENESS TO ALLOW FOR ENEMIES TO PLAY.

LOADOUT


Spy

Ammo packs along the map as well as alcoves, corners and cover so that the class can uncloak and engage nearby the enemies is important. Too open areas will not allow enough time for enemies to be fooled for the class to get a pick and retreat.



CLASS - SPY

CLASS STATS
HEALTH - 125 HP [185 OVERHEAL]
SPEED - 107% [320 Hu/s]

CLASS INFORMATION
AN INFILTRATION CLASS WITH THE ABILITY TO GO INVISIBLE AND DISGUISE AS ENEMIES AND A MELEE WEAPON THAT INSTANTLY KILLS FROM BEHIND. THIS CLASS HAS A LOW HEALTHPOOL AND SEVEN PERCENT INCREASE FROM THE BASE SPEED. SPY MUST UTILISE HIS TOOLS TO GET BEHIND ENEMIES AND ELIMINATE TARGETS. ADDITIONALLY, SPY HAS THE ABILITY TO USE HIS ELECTRO-SAPPER TO DISABLE AND DESTROY ENGINEER BUILDINGS.

LOADOUT

TO CATER THE LEVEL TO THIS CLASS, IMPLEMENTATIONS OF EFFECTIVE FLANK ROUTES, AMMO PICK UPS ALONG THEM AS WELL AS COVER HELP. FLANKS AND AMMO ALLOW SPY TO GET BEHIND AND STAY INVISIBLE FOR LONGER. COVER SUCH AS ALCOVES AND CORNERS ALLOW SPY TO UNCLOAK AND QUICKLY GET A PICK ON A TARGET.

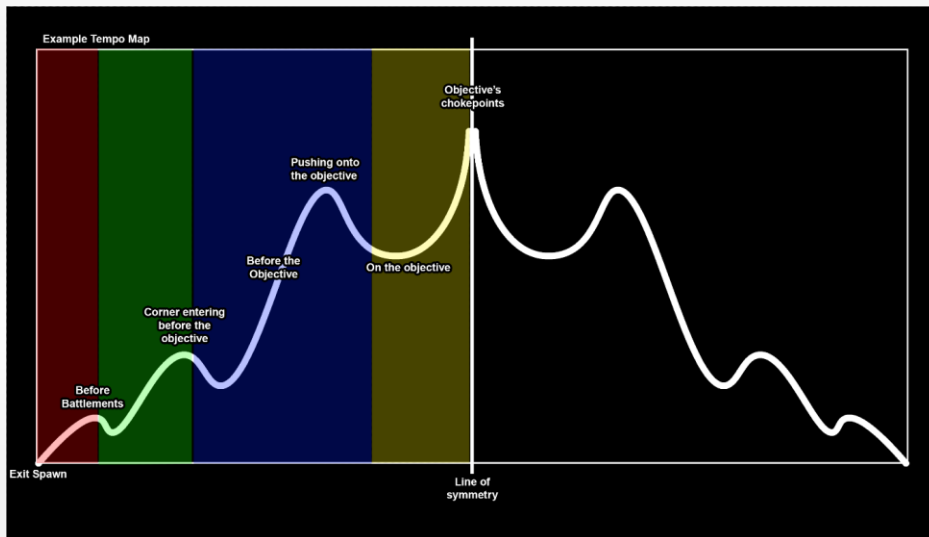
King of the Hill Considerations

As well as considerations for gameplay, further considerations for the level such as consideration of the elements necessary for King of the Hill map as well as additional conventions found within Team Fortress 2.

Timings and Tempo



Level timings for King of the Hill maps are important as to gauge the conventions within a level and the time it takes for a player to return to the objective. For Team Fortress 2 and the gamemode that is relatively quick. Timings that are posted here are just to gauge the amount of time it takes to get back and have the level within a similar timing. The level should avoid timings from being too high or low and in line with the gamemode's average times. All these timings are done as the Heavy class, he is the slowest and therefore will have the lowest margin of error when timing. The tempo of a typical KOTH level can be seen below; the conflict points and the intensity of the average map can be found below as reference for the considerations of the tempo of my level.



Name of Level	Spawn Area Timing	Objective Timing	Total
Badlands	12.76s	4.03s	16.79s
Lakeside	11.60s	5.27s	16.87s
Nucleus	6.99s	4.03s	12.61s
Sawmill	13.63s	7.23s	20.86s
Viaduct	13.28s	2.88s	16.17s
Average Time	11.65s	4.69s	16.66s
Range	6.99-13.63	2.88-7.23	12.61-20.86

Level Conventions and Layout



Within every game there are necessary components that are used within gameplay. These components can broadly be considered as the conventions within a game and consist of design and gamemode considerations to interactive objects. Within TF2 there are a checklist of conventions within the level that is to be included in the designs as they're either incredibly important for the balance and enjoyment of the level or required to function. The checklist of inclusions can be found below.

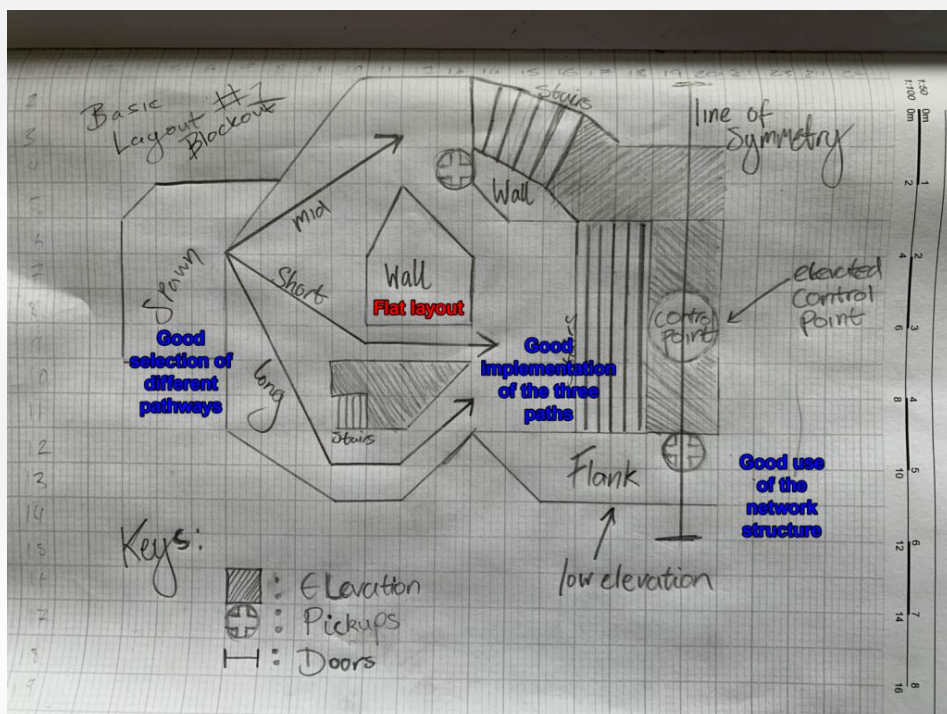
- Valid spawnroom equipped with the capabilities of resupplying players, changing classes and respawning.
- Spawnroom also uses blocking volumes to prevent enemy players from entering the enemy spawn.
- Functional objective which is capturable and counts down a timer which once finished announces a winner and the round is reset
- Pickups within the level that allow players to replenish their health and ammo
- Functional skybox to prevent leaks and obvious visual errors.
- Level should be symmetrical, and each side balanced with a focus on the objective being central

Map Overview

Map Structure

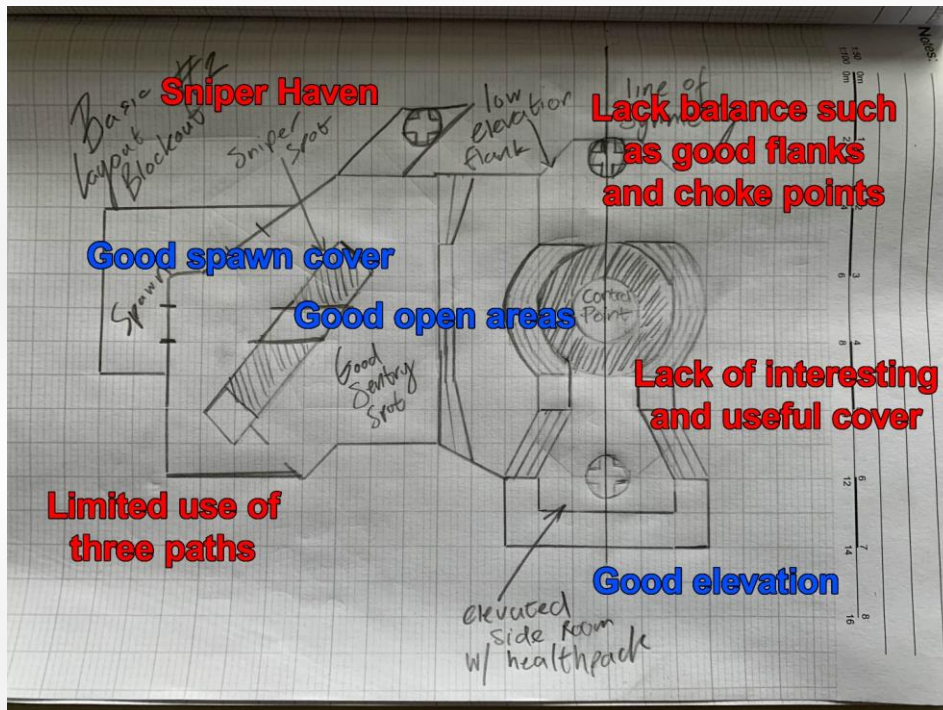
Level Structure

The first approach was focused on having elevation on the point and creating player pathing which worked for each role. A short for classes like the heavy, with choke points for power classes, open areas and flanks for pick classes. Class and playstyle considerations were a focus on this design, but it's lacking in terms of elevation and structural conventions of TF2. For the final design of the level, this iteration was used as a basis for the overall level layout and several conventions.

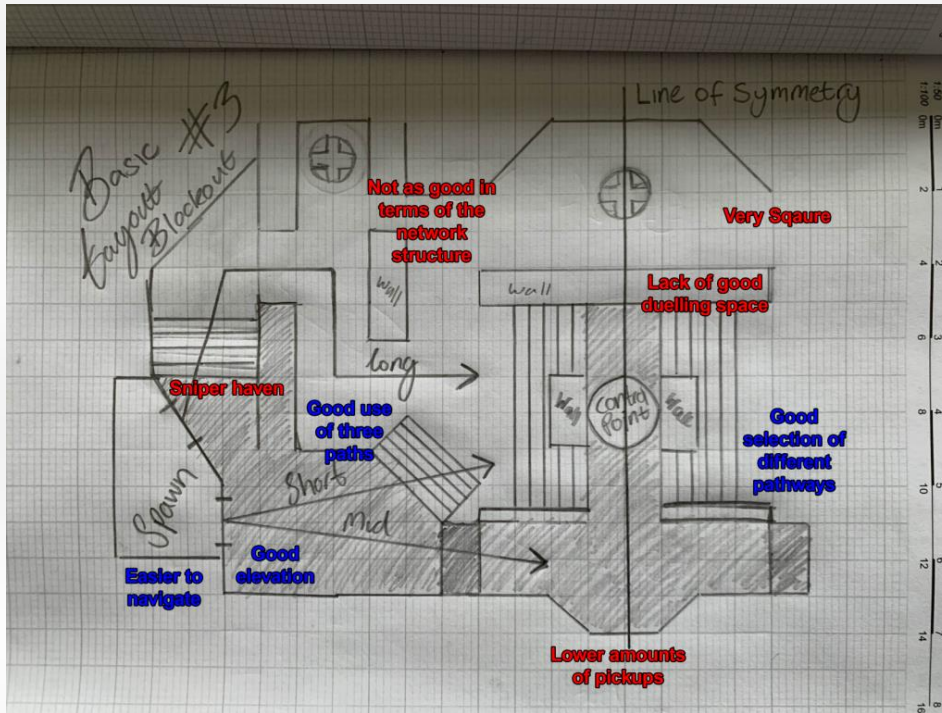




For the second design, the focus was on elevation whilst implementing the balance for each playstyle considering its implementation. Because of this it didn't account for the poor sightline balance and lack of effective flanks. As a result, the level for this design seems weaker in terms balance but the multiple levels of elevation are a consideration implemented into the final level's design.



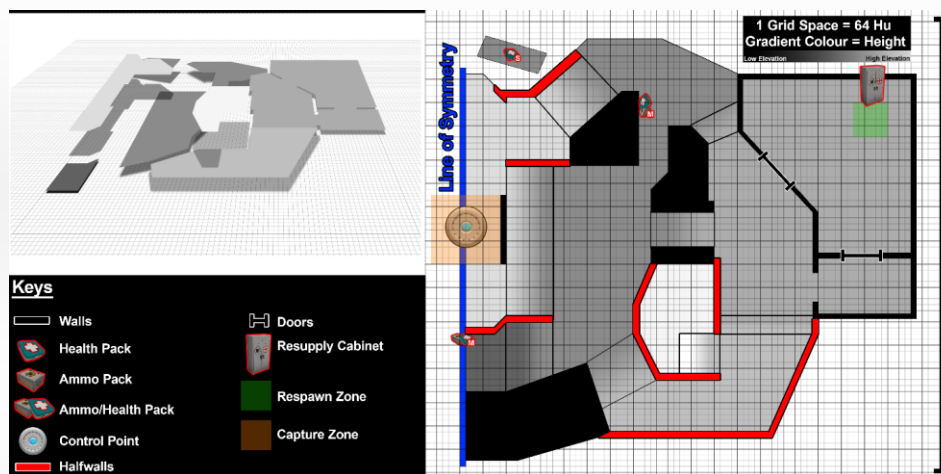
The last design's focus is on creating a level that conforms to the square design of Team Fortress 2. The approach was to include a variation of height, level conventions and improve on the lack of effective flank routes. As a result, many of the same issues present in the previous iteration do not appear. But issues such as the sightlines exist. The level's square design is very navigable and would serve as a design consideration for plotting out the final design iteration.





Improved Design

The improved design is based on the first initial design with the updated inclusion of further considerations. Refining on the aspects in the initial designs and implementation of conventions (healthpack, respawn point, doors, etc) were the additions for this iteration. The basis for the level comes from the first initial design with elevation and the square structure considerations included from the others. Everywhere outside of the level is a deathpit with a steep drop to the player's death.

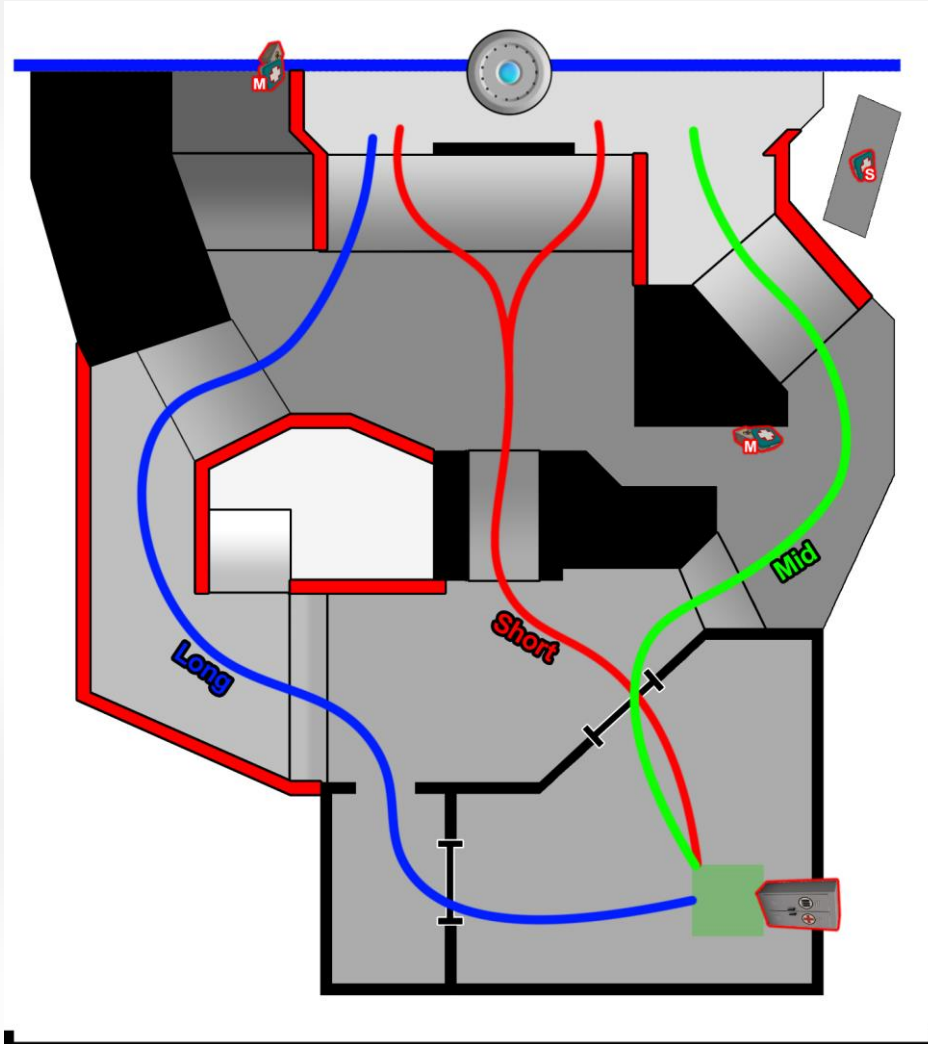


Three-Paths

The three paths in the level are focused on pushing certain playstyles and characters into certain directions. The goal of which is to help push specific interactions and provide alternative routes around the level. The broken-down overview of the three-paths in the level can be seen below. The mid route is designed to be for power classes, the frontline of defence and is the second fastest route to the objective. Power classes can take this route, but it would be longer than the short route. The short route is the path of least resistance and is a good way of progressing towards the objective. It's intended for classes such as Scout and Engineer as quick but slightly defensive areas for teams to fall back on. The long route is designed for pick roles and to provided weaker teams alternative routes around the enemy. Since it heads towards and flank and there is an optimal sniping position pick classes are encouraged to use this route. Additionally, the larger distance away from the objective and the intended power heavy mid route



could help disadvantage teams to not be held back as much.

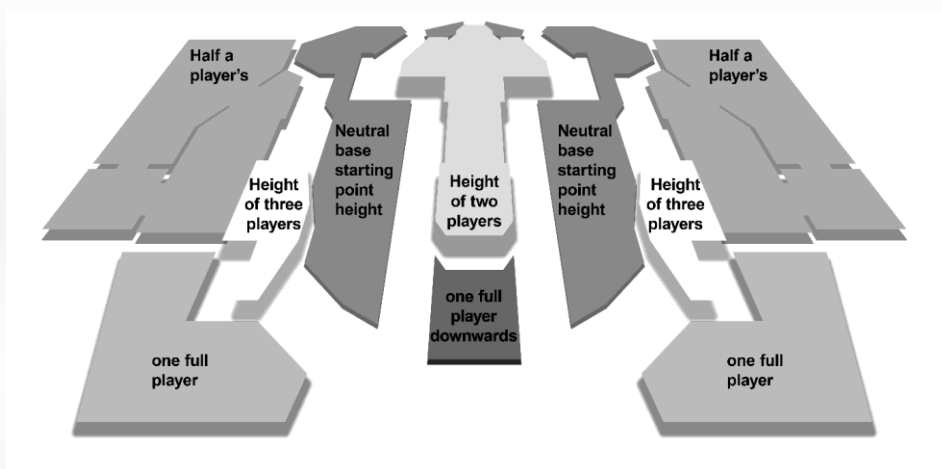


Elevation

Elevations within the level has been considered and detailed on a map. The elevation consideration is a rough estimate of what each portion of the level's height should be within the engine. By having these rough estimates, the level is more detailed by

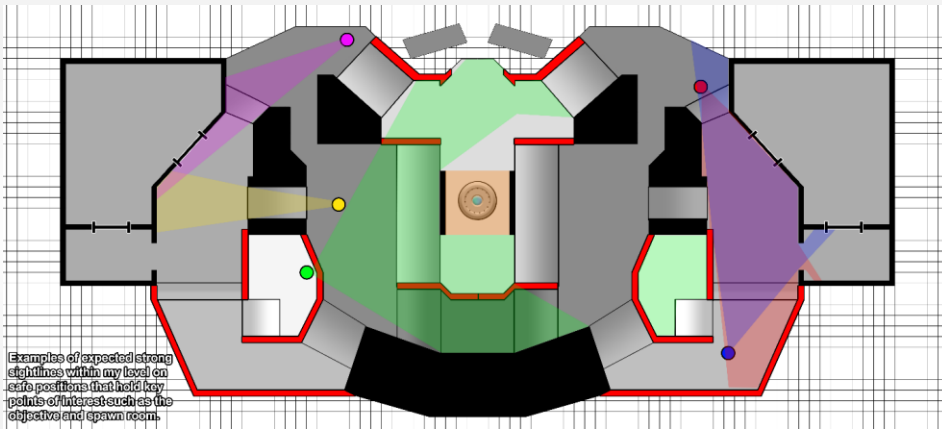


showcasing intended elevation at certain points. The main two classes for elevation are Sniper and Soldier for sightlines and rocket jumping respectfully. Increased elevation can help create safer sightlines and slightly more advantageous positions for teams. The intended implementation of elevation for this level is to create an elevated point which can be held by the winning team and block major sightlines for losing ones. Elevation is slightly increased in the spawn areas to allow teams that are pushing out from there at a slight advantage and a single spawn sniping spot far less effective by making the shot's head height change.



Sightlines

Sightlines are massive consideration for the level because the effectiveness and frustration that can occur when the class Sniper is powerful. Overall, the sightlines in the level should look like this for the major positions that a Sniper would hold look like below. Additionally, in the initial design I made consideration for the class by keeping the safe protected and I have intended to continue that with my second design. Sightlines have been tested on the design and considered in the iteration process. This is to ensure that sightlines are not too strong on the objective and the spawn is safe from being effectively camped by a sniper.

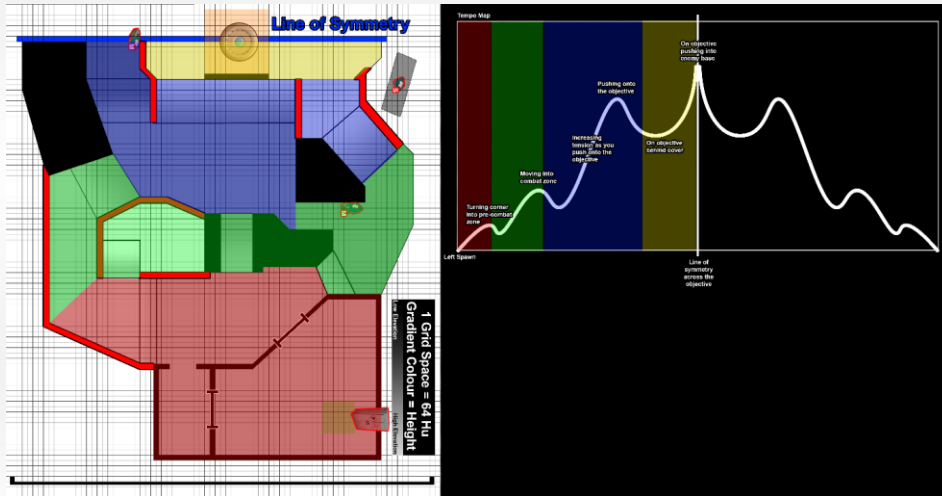


Timing and tempo showcase the intended flow of the level by displaying the intensity of specific areas alongside the time to reach them. Timing and tempo are connected so a smaller, lower timed map can compensate with a high tense and compact level where teams have difficulty pushing forward. For the level it's important maintain a steady flow with an intended medium tempo alongside medium timings lengths.

Level tempo is broken up into key areas and how tension rises and how the level is expected to change throughout the level. The high peak at the centre for the intended level tempo is to prevent pushes from being too powerful hence the two large tension and how the level is built around maintaining the tension. Tension raises when players



get closer to the objective with areas behind the main combat zone being relatively safe.



Level Timings

Throughout the development phases maintaining level timings supports effective implementation of tempo changes throughout the stage that would affect it. To measure, heavy is used as a basis with the slower speed giving more precise data. Ideally, the higher tension throughout my level should have timings relatively low, high times should be avoided due to the tempo. The overall iterations have no yielded substantial changes to the level timings and doesn't affect the tempo.

Name of Iteration	Leave team side	Objective	Total
Blockout	6.98s	4.80	11.79
Third Iteration	8.35s	4.51	12.87

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Blockout

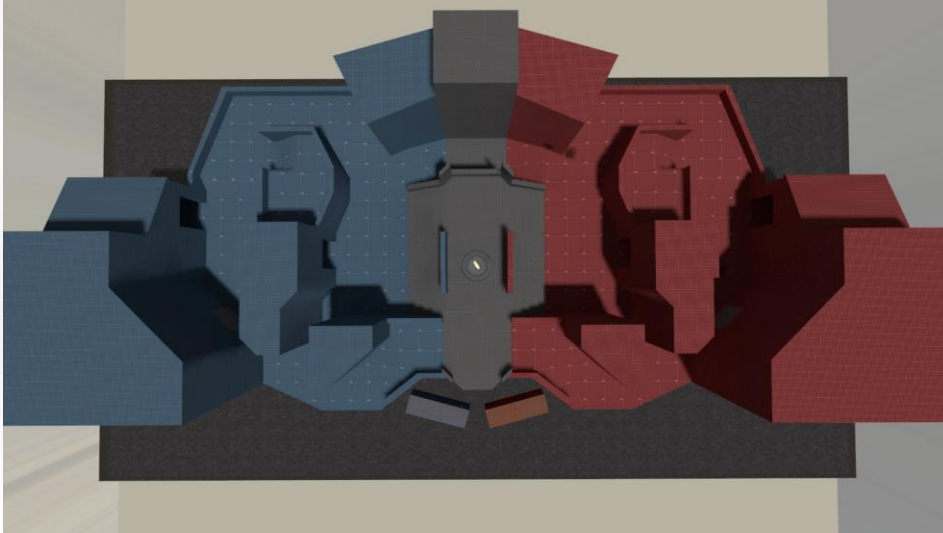
Initial Blockout

The initial blockout of the level is focused on building on the designs. Having an effective basis will serve well in allowing focus on smaller nuances within the level such as visuals and smaller interactions instead of larger sweeping changes. Overall, the level achieved this, creating a well-balanced and good experience that is enjoyable to

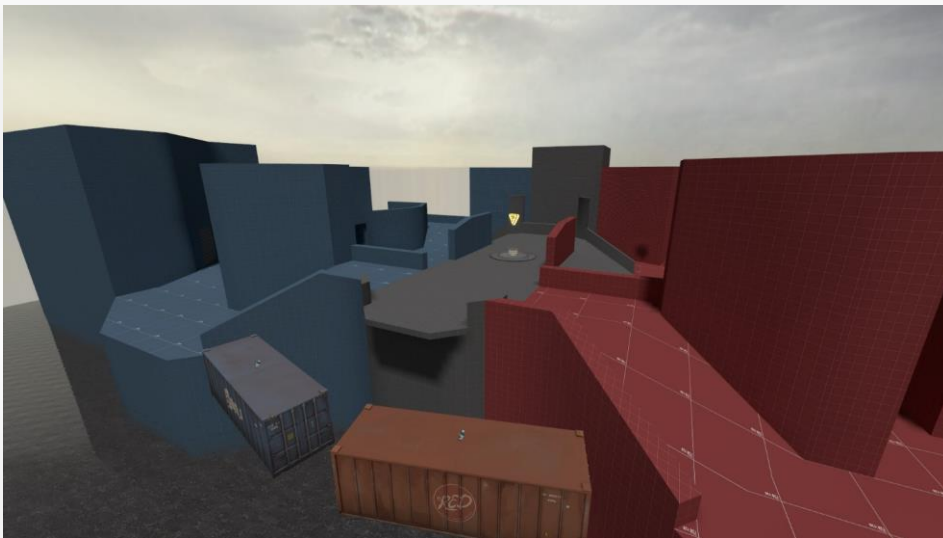


move around and play within. Several early renders can be seen below showcasing the before and after of the many changes from the original design.

Over-head view



Map Camera





Level Final Iteration

Point overview



Changed the over the water route to improve usability and reinforce more affordant design. Visual improvements have also been made as well as the addition of ammo packs to assist spy and engineers who might utilise this path. The objective has been updated to be approachable, defendable and to create a tighter chokepoint. Addition of player clips have been added to improve player interaction and avoid weird movement related issues such as getting stuck on geometry.

Mid path



Changes have been to add walls to reduce the chances of player being knocked off utilising this route as well as the inclusion of signposting. Minor changes in line with the other areas.

Area before control point



Inclusion of props to help with affordances and add additional paths improve players perception of this area. Affordances highlight key areas for the player and reinforce the sturdy perception of the environment.

Long route and sniper battlement



The addition of affordances helps to draw attention to the route and the key elements on the pathway. The box was added to help improve cover and offset negative shape theory. The sniper battlement's cover has been reduced to reduce camping and expose snipers more.

Spawn exit



Affordances have been used to highlight the key areas of spawn exits. This will help players focus in on the important portion of this structure in the area. Design remains the same besides affordances.

Inside spawn room



Spawn room utilises negative affordances alongside positive affordances to highlight the exit and resupply locker. The main spawn room has been pushed back to improve player safety whilst respawning.

Flank route



Flank route has used affordances to highlight the pathway and to better showcase the location of the pickups within the area. The overall design has not been changed, only the inclusion of affordances to improve player behaviour.

Short path



Pathways are clearly highlighted with signs to support player behaviour and clearly mark the options in pathways. Lighting and shape affordance in the short path are used to reinforce shape theory and highlight the pathway.

Over-head view



Layout changes have remained similar to the main designs with slight variations to improve player interaction and behaviour. Visually, the level has been improved from the original blockout which is distinct and based on pre-existing visual considerations complete with textures and props.

Map Camera



Additions of landmarks help to highlight key areas and orientate players to the main objective with the large obvious signs and flora. The gameplay camera is balanced between visually appealing the shot of the level and showing the main gameplay area. The timings for the objective have been reduced in response to the low-level timings.

Character Experience

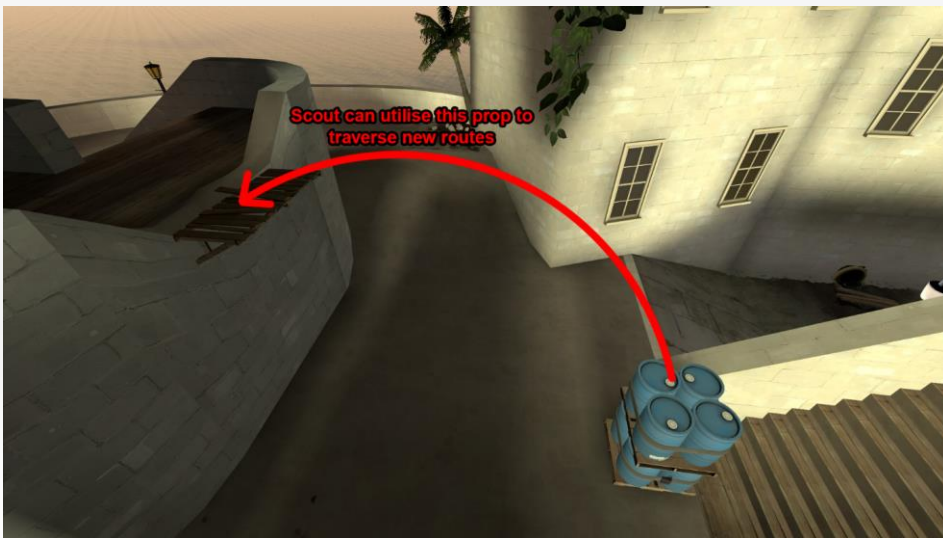
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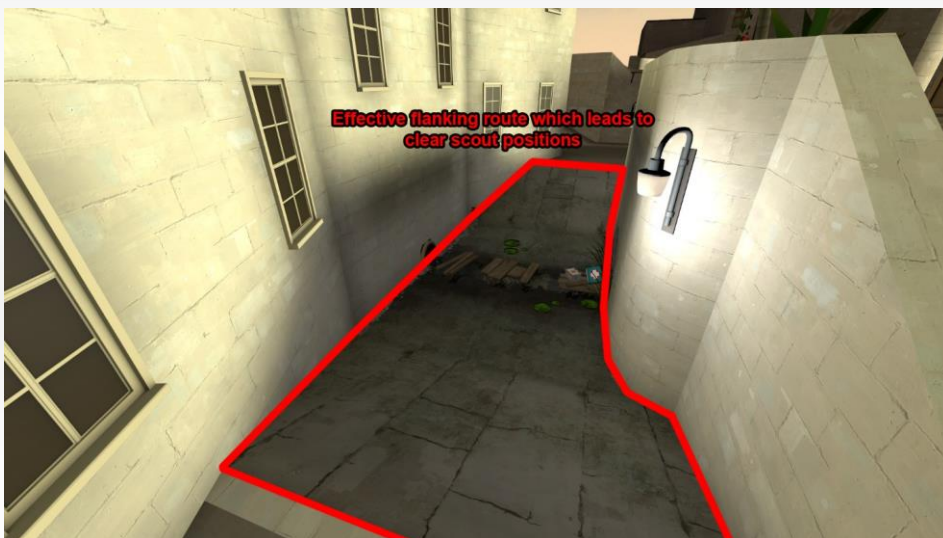


Scout

Specific implementation for Scout has been considered for the level. Implementation for this class comes in the form of additional paths and effective flanks. Flanks for the class are efficient, opening to relatively open areas where the class can utilise his duelling prowess to win fights. The additional consideration of an added route for Scout allows the class to reach the snipers nest by utilising a prop to double jump to reach this area.







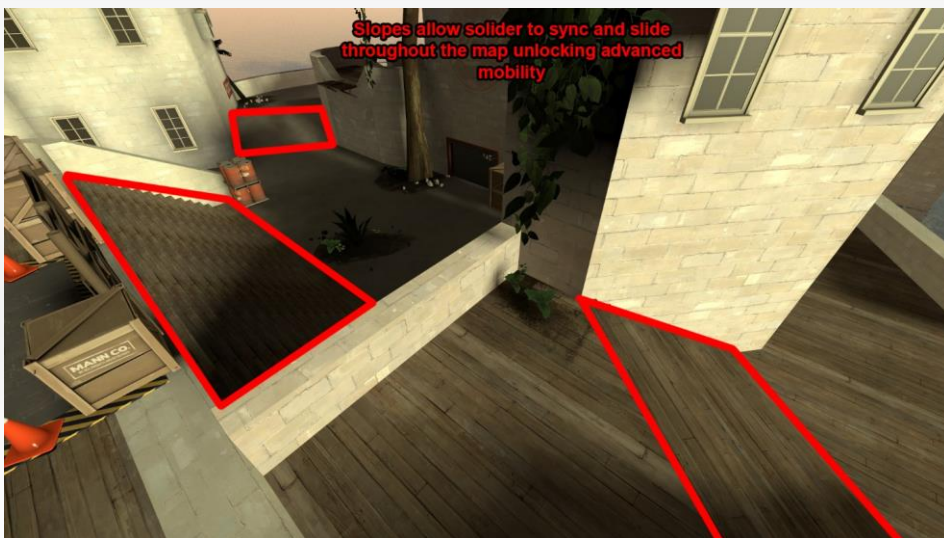
Soldier

Soldier's implementation within the level can be found with the implementation of elevation, more specifically the sloped elevation. Soldier's blast jump allows additional options within the level, the sewer flank and sniper nest are key locations where basic rocket jump techniques can be used to effectively take advantage of this mechanic. Additional implementations of more advanced forms of rocket jumping can be found



in the level with implementation of slopes. These slopes work well with the physics in the engine allowing the class to sync up rocket jumps for faster level travel. These two implementations add options for the class for all skill levels.







Pyro

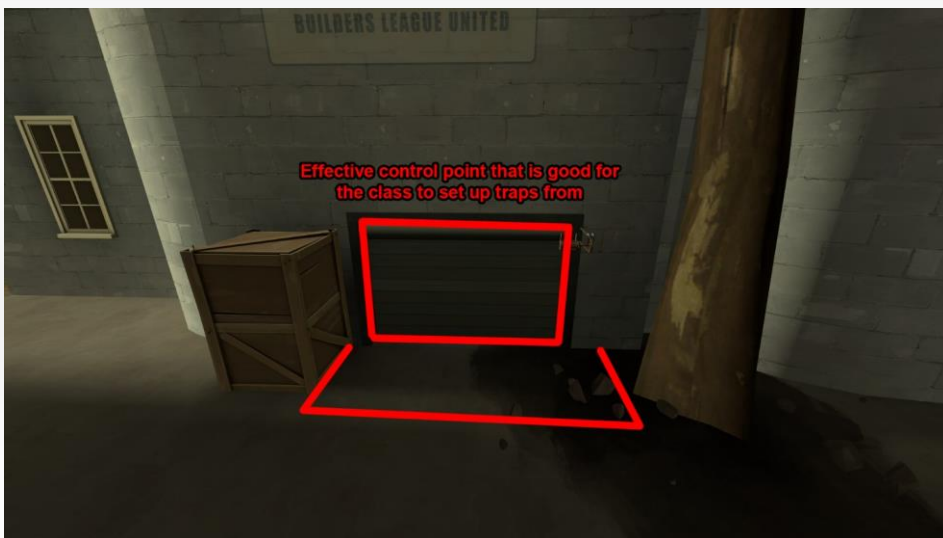
Implementations for Pyro are in the form of an ambush hallways which the class can take advantage of. In addition to that the implementation of a large environment hazard in the form the water below allows the class to knock enemies to quick and effective death. Whilst the map might not be heavily suited in every way these concessions offer the class greater heights.





Demoman

Implementation for Demoman are the inclusions of effective chokepoints and a semi-enclosed objective which can effectively be held by the class. Chokepoints help the class to funnel enemies into effective position for kills whereas the objective utilise walls which create further chokepoints which this defensive powerhouse can benefit from.





Heavy

Implementation of strong cover and a medium sized area which the class can defend are good for the class. There are several in the level that meet those specific



requirements for the class such as the pathway from the short and mid paths to the objective which have plenty of cover and effective for the class to approach/defend.



Engineer

Implementation of engineer have been to give good positional locations for the class to place down buildings. Firstly, effective strong cover alongside safe positions where



teammates can teleport in and resupply by the dispenser. Good sentry spots are also a must, and engineer should have a good position for his gun which is hard to shoot at and covers a decent area.







Medic

Plenty of effective cover has been implemented for the class to offer greater safety. The level is quite covered in the design, but additional considerations have been made for the class to improve playability and that is found in the form of added cover. Below are some examples of cover added to help protect the class in certain approaches, allowing the class better opportunities to cross sightlines.





Sniper

Sniper implementation is less about improving the classes viability but instead focusing on creating balance and awareness of key sniper locations. Within the level several helpful design considerations have been added to manage the sightlines and make the experience of dealing with this class easier. Firstly, implementation of affordances in the form of landmarks help to focus the attention on a key sniper location. Cover on the



objective can be looked through and players can be aware of a sniper location. Cover has also been placed around several points to allow players to weave in and out of the key sightline. The best position is flankable and it's up to the sniper to be aware of his surroundings or be easy pickings for pick classes like Scout or a Spy.



Spy



Plenty of cover to decloak from, flank routes and ammo packs help to get the class into a good position. The over the water flank is a perfect example of a flank design for the class. In addition to that the additional cover placed acts as good places for the class to initiate their attack sequence through decloaking and attempting to backstab.



Points of Interest
Pickups



Pickup's implementation within the level have been focused on acting as affordances and supporting players pushing objectives. Firstly, the placement of pickups has been specifically implemented to support the tempo of the level by creating defensive holds in locations before the objective. Due to these placement considerations, players pushing have a slight advantage over the objective do overcome the imbalance in elevation from the defender holding the objective.



In addition to this, pickups act as breadcrumbs for the flank routes and are placed to encourage players to flank more often.



Affordances for these elements have made sure to highlight their key implementation areas either with props or overlays. In addition to the base affordances in one location the implementation of lights is used to draw attention towards these props. A good example of affordant implementation can be found below.



Control Point

Implementation of the control point within the level is focused on creating defensible space whilst not creating an impenetrable hold. The implementation of the cover on the objective prevents easy attacks to players who are on the objective by being outside of most attacks. Whilst the chokey nature of the control point and multiple entry points means that effective defensives rely on players countering pushes.



Affordances for the control point include the implementations of the visual elements of the objective which showcases the current objective holder and location of the capture point. Alongside this the addition of clear outlines for the objective help to highlight the capture area properly for players.



Spawnpoints

The addition of spawnrooms within the level have been focused on using several affordances to improve recognisability and navigation of the spawnroom. For the outside, the position and overlays help to highlight the spawn room. The clear use of affordances on the door help to create identifiable locations for where players will exit.



The implementation within the spawnroom utilise focal points in the form a single exit which focuses on the actual exits from spawn. In addition to that, the use of lighting affordances highlights the doors within the room and allow players to identify these exits. The simple layout of the room with the clear affordances makes it difficult for players to get lost within the space.



Balance

Sniper Balance

Sightlines have been a core consideration throughout the development and update cycle of the level. Sightlines have been balanced between giving opponents opportunities to counter and react to the class. Firstly, options players have to counter the sniper come in the form of effective flanks and routes for the classes. Flanks are



spread out enough that the class will be unable to hold both positions and must be aware of their surroundings.



Secondly, the reduction in self-cover for the class means that enemies are able to more effectively hit the class due to them being more in the open. Not massively being out of the way helps to balance the class from being too safe.





In addition to that, the implementations of affordances and transparent covers allow classes on the objective or moving forward to recognise the class and react. The large landmark of the tree helps to draw focus whereas the cover on the objective can be used to see if a sniper is holding a specific angle.



Overall, this mixture of safety, power and cover help to not disincentives the class as the character still has a relatively effective position. But instead aims to create more engaging gameplay by creating interactions with the class and allowing players to react.

Engineer Balance

Engineer is the next and last class to specifically discuss within the design and balance of the level. Firstly, teleporters aren't as useful here as the short level and lack of brilliant positions means that aspect isn't incredibly strong. So, the balance for that isn't an issue.

As sentries, the optimal placements of the buildings within the map are balanced between the area they cover and safety of the sentry. The position on the point and sniper nest are strong, elevated positions that cover good space within the map. The more covered and further placed back sentry has more cover but covers less space. This creates good balance for the class that results in the class becoming weaker the more aggressive the position and allows for the class to have variety by having multiple good options with downsides.





Dispenser placement has been considered, to optimal positions are the objective and the mid route. The position of the dispenser on the mid route is no more effective than the healthpack in this area and allows the same balance between pushing the objective and healing mentioned before.



As for the objective the chokey nature of the point will result in the dispenser being killed in large scale pushes so the building isn't as safe. Whilst a strong position only teams that are dominating would hold it, with the same risk and reward as sentry with more cover within the level.



Flanking Balance

Flanks within the stage have been balanced to give opportunities to teams when pushing onto the objective. Whilst the defending team can hold both flanks it will result



in the teams spreading the power evenly reducing the defensive hold on the objective.



The limit of two flanks means that players are consistent from where they arrive from so players can be cautious of the routes that players might come from. This reduces a good player's chances from dying randomly to players they cannot see and weren't able to predict due to the too many routes.

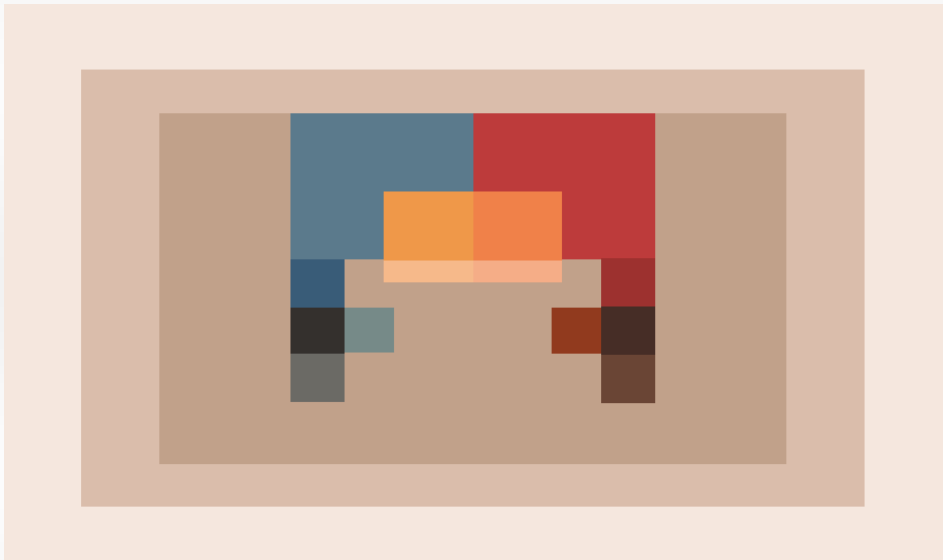


Aesthetics

Visual Development

TF2 Visual Style

The conventions of level design must conform also with the visual cohesion of the rest of the game. TF2's art style is specifically unique, utilising art styles from mid 20th century comic artists from three artists: J. C. Leyendecker, Dean Cornwell and Norman Rockwell. Additionally, Valve has discussed their approach to levels and their visual implementation. Firstly, the obvious inclusion of the visually contrasting triadic colours of red, blue and yellow. The colour palette for the game can be found online and was created by Valve to showcase the intended implementation of colours for items and levels. Secondly, the colours used for levels should ideally be desaturated with the more saturated implementations for specific key elements of highlights. Simple walls should utilise desaturated colours whereas implementation of a spawn or pickup should use more saturated elements. This is done to implement less visual noise and not take away from the distinctive colours and silhouettes of the playable classes improving readability.



Overall, the level should have minimal levels of visual noise in areas of play to avoid drawing attention away from gameplay. For each team there are specific implementation for visual implementation of structures and textures. Ideally, the RED team focuses on utilising wooden, warm textures with structural elements that implement angular implementations. Whereas the BLU team utilises cooler colours and

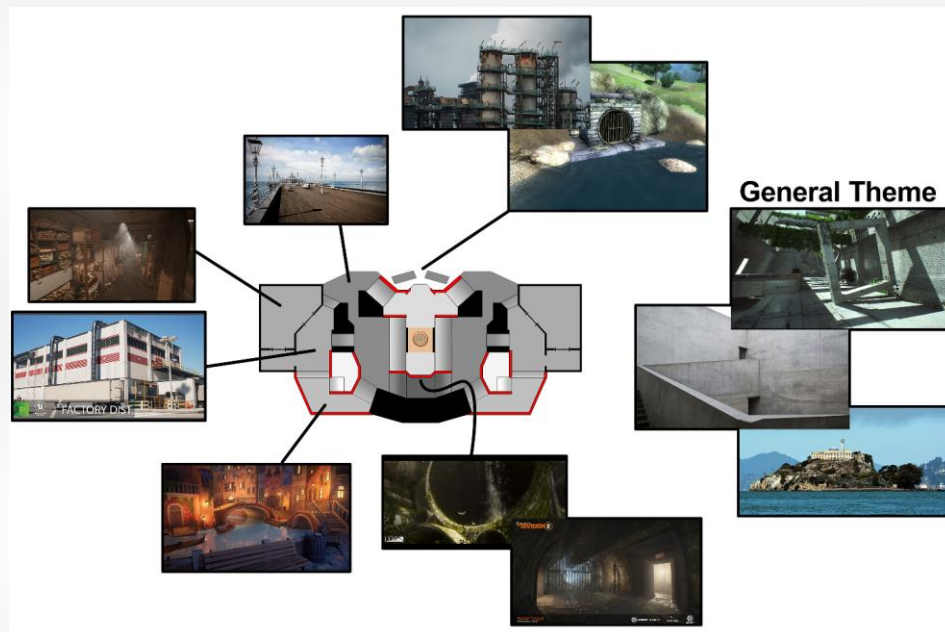


industrial textures such as stone and metal; structure utilise orthogonal forms. A link to the reference on TF2's art style can be found here in the PDF below. This is the approach to the level's visual implementation and the basis of all future developments of this level visually.

https://steamcdn-a.akamaihd.net/apps/valve/2007/NPAR07_IllustrativeRenderingInTeamFortress2.pdf

Level Visual Consideration

Alongside other designs, visual level consideration focuses on the visual look of several of the areas within the level and the ideal objective for how each section should look like. This section will be the showcase an explicit explanation of each section and their idea visual implementation. This part of the LDD will help to bridge designers and environment artists together.

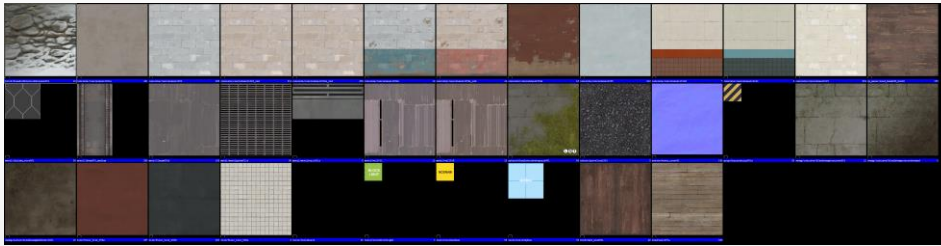


Visual Development

The visual development of the level has been focused on applying the above visual considerations within the conventional visuals of the art style. The level focused on utilising textures that are desaturated to help players differentiate against the



background. These textures are in line with the concrete structure theme with red and blue side textures alongside neutral colour textures. This helps to create clear areas for each team which are reinforced with affordances. Other textures throughout the level have either utilised this theme or are used neutrally throughout the level such as in the case of the wooden plank flooring texture.



As for the props that have been used have been focused on implementing the art style, for example the implementation of Italian style lampposts and tropical flora have been to progress on the ideas of the specific area's visual consideration or the general theme with the level. The visual noise of the map has been kept at a low level to not distract players and so props that are only visual have been placed outside of the levels bounds to not create visual imbalance or distract players.



Affordance

Affordances Overview

Commented [DB2]: What affordance techniques have been considered to help convey information and direct player behaviour? (Remove this comment before hand in)

Commented [DB3]: What elements need to be considered to ensure the map is balanced? (Remove this comment before hand in)



Affordances have been used throughout the level in multiple different methods. For this level, affordances have been used everywhere to influence behaviours and push players in specific directions. Several affordances are used in conjunction, near or around each other, usually with the purpose of encouraging these implicit behaviours.

Light Affordance

Light affordance is using lighting to push players behaviours. By creating lighting to positively and negatively you can influence players in directions with higher lights encouraging players and vice versa. In the level several instances of light affordance are added to influence player behaviour; some examples being the spawn room, pick up location and flank route. For these examples, the implementation directs players behaviours towards certain directions and conventions such as leaving spawn and picking up healthpacks.







Shape Affordances

Shapes can be used implicitly to reinforce player perception. Shape affordance is used twice in the level, specifically the square shape theory to reinforce the idea of sturdy, hard structures. These examples can be found specifically on the long and short route which utilise shape theory to create that affordance. As such these areas now will be perceived differently.



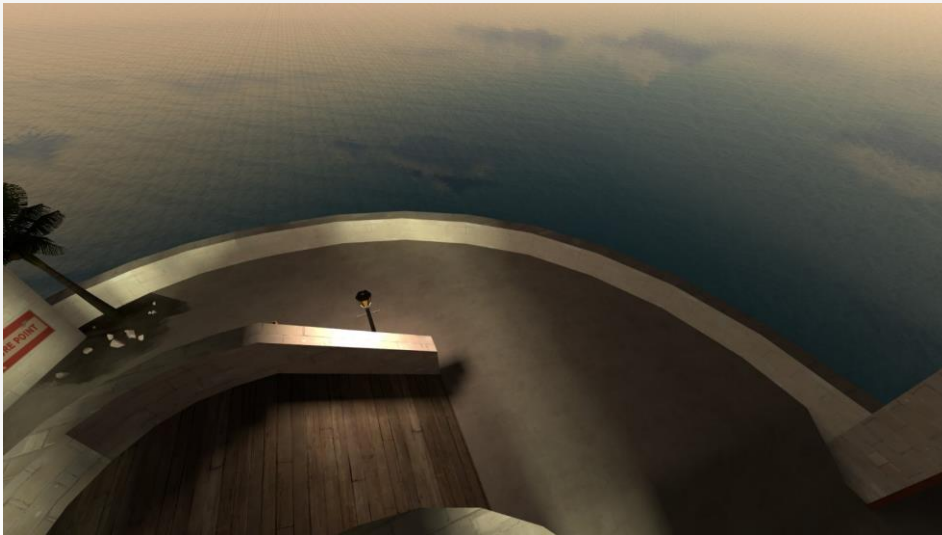


Leading Lines

Leading lines are another type of player directing affordance which utilises implicitly found lines to direct player behaviour in the direction of the line. Leading lines are used within the level to direct players along the long route by utilising a rounded line pointing



towards the objective. Implementation within the level support players freedoms and encourages players to utilise the path more as it makes the path more recognisably easier to follow.



Colour Affordance

Colour affordances have been used throughout the level to create recognisable team-specific regions which are in line with the conventional approach of TF2's team-sided level design. Colour theory has also been utilised to highlight and contrast regions within the level such as the boundaries, doorways and objective. Colour theory in this instance creates contrast which can be used to implicitly highlight ideas within the level, all of which the use cases achieve. The examples showcase the myriads of instances this affordance has been utilised to create better implicitly designed levels.





Breadcrumbs

Breadcrumbs have been utilised in a limited capacity within the level to direct player behaviour alongside the flank routes in my level. Two primary examples of breadcrumbing within the stage can be seen in a minor and major capacity. In the latter example, smaller pickups have been used to help direct players along this route whereas the former an obvious health pack will incentivise players to take this route



more often. This implementation will encourage player choices and help lead players along certain, less visible paths.



Landmarks

Landmarks are obvious, clear indicators that are present throughout a level that can be used as reference for orientating one-self with the map. In the examples provided the



large props are clear and visible in the centre of the level which support player orientation. This will decrease the chances that players will find themselves lost or unaware of where enemies and teammates would appear from.







Signposting

An explicit form of affordance, signposting is used to directly tell players that certain key elements are along specific highlighted paths. On each team-side of the level, this affordance is included to highlight the potential routes towards the objective. Each sign is clearly visible and can be seen once leaving out of either exit within the spawn room.



This implementation was to support player choice and allow players to be aware of specific routes.



Anti-Affordances

Anti-affordances are implementation of affordances with the goal of reducing the approachability of specific elements. For the level, anti-affordances are used to support



players by creating undesirable space by creating unreasonable areas to explore. Two areas where negative affordances are used are found within the spawns and with the height of buildings. Firstly, the height of buildings is set much higher than an average blast jump making reaching them relatively unreasonable. This should disincentivise players from attempting to approach from these blocked off areas. Secondly, within the spawn, props are placed outside of clear paths towards the door to discourage players from wasting time by creating negative space.





Testing

Testing Overview

The stages of testing supported the development of changes within the level. The general overview of feedback ranged in the positive to highly positive range with responds asking for smaller changes. The volunteers in testing varied in skill-level giving



the feedback merit in terms of design as I had larger pool of differing opinions. The small sample size was an issue throughout testing was an issue and was considered for changes. Each individual stage asks participants specific questions about the level which focused on aspects related to the iteration.

- The first iteration focuses on asking about balance, navigation and class dynamics which was met with positive reception. Minor changes were made in response. Generic rating for visuals and design as well as asking about exploits/issues.
- The second iteration aspects of testing were about affordances and balance, with further questions focused on specific navigation elements. Minor changes were made in response to the feedback based on positive reception in responses. Generic rating for visuals and design as well as asking about exploits/issues.
- The third iteration focus on feedback such as visuals and issues with the current visuals. Additional questions about balance with minor changes made with a positive reception to the changes made. Generic rating for visuals and design as well as asking about exploits/issues.
- Final iteration asks about exploitation and general class balance. Met with positive responses, no changes made in response due to lack of effective information to iterate on.